

PROCEEDINGS OF SPIE

SPIDigitalLibrary.org/conference-proceedings-of-spie

Portable flow device using Fourier ptychography microscopy and deep learning for detection in biosignatures

Nguyen, Thanh, Aslam, Shahid, Bower, Dina, Eigenbrode, Jennifer, Gorius, Nicolas, et al.

Thanh Nguyen, Shahid Aslam, Dina Bower, Jennifer L. Eigenbrode, Nicolas Gorius, Tilak Hewagama, Laddawan Miko, George Nehmetallah, "Portable flow device using Fourier ptychography microscopy and deep learning for detection in biosignatures," Proc. SPIE 11401, Real-Time Image Processing and Deep Learning 2020, 114010H (22 April 2020); doi: 10.1117/12.2557316

SPIE.

Event: SPIE Defense + Commercial Sensing, 2020, Online Only, California, United States

Portable flow device using Fourier ptychography microscopy and deep learning for detection of biosignatures

Thanh Nguyen^{a, c}, Shahid Aslam^{a, *}, Dina Bower^{a, b}, Jennifer L. Eigenbrode^a, Nicolas Gorius^{a, c}
Tilak Hewagama^{a, b}, Laddawan Miko^a, George Nehmetallah^{c, #}

^a NASA, Goddard Space Flight Center, 8800 Greenbelt Rd., Greenbelt, MD 20771, USA;

^b University of Maryland, College Park, MD 20742, USA; ^c The Catholic University of America,
620 Michigan Avenue NE, Washington DC 20064, USA

*shahid.aslam-1@nasa.gov; #nehmetallah@cua.edu

ABSTRACT

A proof-of-concept, compact, portable Fourier Ptychographic Microscope (FPM) to perform wide field-of-view, high spatial resolution imaging ($<1\ \mu\text{m}$) for biosignature motility in liquid samples, is presented. The FPM has the potential method to be developed as a space-based payload for future landers destined to the Ocean Worlds. A portable FPM using an existing Fourier ptychography (FP) algorithm adapted for reconstruction is demonstrated. A NVIDIA Jetson Nano board and camera combined with FP, is used to computationally reconstruct sub-micron resolution images. Additionally, deep learning was employed to perform inferencing prediction which enables the on-edge FPM device.

Keywords: Fourier ptychography microscope, artificial intelligence, deep learning, biosignatures

1. INTRODUCTION

Over the last 25 years, there have been rapid advances in microscope technology enabling high resolution imaging in a variety of small and automated platforms [1, 2] for applications in the biology, geoscience, and materials science fields. Although not used specifically for life detection, optical microscopes have been included as research instrumentation in planetary research missions. A high-resolution imaging system was developed for Curiosity, MAHLI [3], and optical microscopes were flown on the Beagle 2 Lander [4] and on the Rosetta Philae mission [5, 6]. On Philae a near infra-red hyperspectral imager, was coupled to an optical microscope. A microscope-spectrometer was developed and launched for the Phobos-Grunt mission [7]. A science priority for planetary exploration in the next decade is to search for potential biosignatures in Ocean Worlds. In this pursuit, it is important to determine what constitutes a biosignature across multiple spatial scales, and how this compares with abiosignatures formed by nonliving processes [8]. Several different microscope technologies can be adapted to resolve and characterize microscale biosignatures in Ocean World samples; these include visible microscopy [1, 9], deep ultraviolet microscopy [10], flow cytometry [11], atomic force microscopy [12], and scanning electron microscopy [13]. For Ocean Worlds microscopy needs, it is highly desirable to capture high spatial resolution images of particles to identify features $0.2\ \mu\text{m}$ in size and at abundances of 100 cells/cc in a 5-cc volume sample [14]. Implementation options will require a sample processing system to interface with the sample observation technology. A variety of optical processing methodologies are available, including phase change using pressure to convert ice to a liquid, followed by sample concentration. Also, biomineralization reactions between microbial life and the geosphere of a planetary body can result in incorporation of microorganisms into the geological record as body fossils or casts that can appear macroscopic ($>1\ \text{mm}$ in scale). Certain biofabrics such as microbiolites (e.g., stromatolites, thrombolites) and biofilms may also be large enough to be resolved at the $100\ \mu\text{m}$ scale, if not larger depending on the habitat. The observation of macroscopic morphological features could serve as a compelling biosignature, especially when coupled with detailed organic and inorganic chemical analyses [15]. On the path to providing a wide field-of-view, high spatial resolution imaging ($<1\ \mu\text{m}$), for biosignature detection and motility in liquid samples, a compact, portable FPM for planetary exploration to the Ocean Worlds is discussed herein.

2. EXPERIMENTAL SET-UP

The key components of the FPM experimental system set-up are shown in on Fig. 1. These are all controlled by a NVIDIA Jetson nano board with state-of-the-art machine learning processing capability. The camera captures low-resolution bright field and dark field intensity images corresponding to multiple illuminations of each lighting element

in the LED array. Different diffracted orders of light distribution serve as the sub-complexed fields in the 2D Fourier domain and will be stitched using a constrained iterative process to perform high-resolution reconstruction. The position of each LED element will determine where the Fourier transform of each low-resolution image is stitched on the global Fourier domain. The LED positions based on setup coordinates will be digitally calibrated with an optimization method during the iterative process along with FCM to force it to converge.

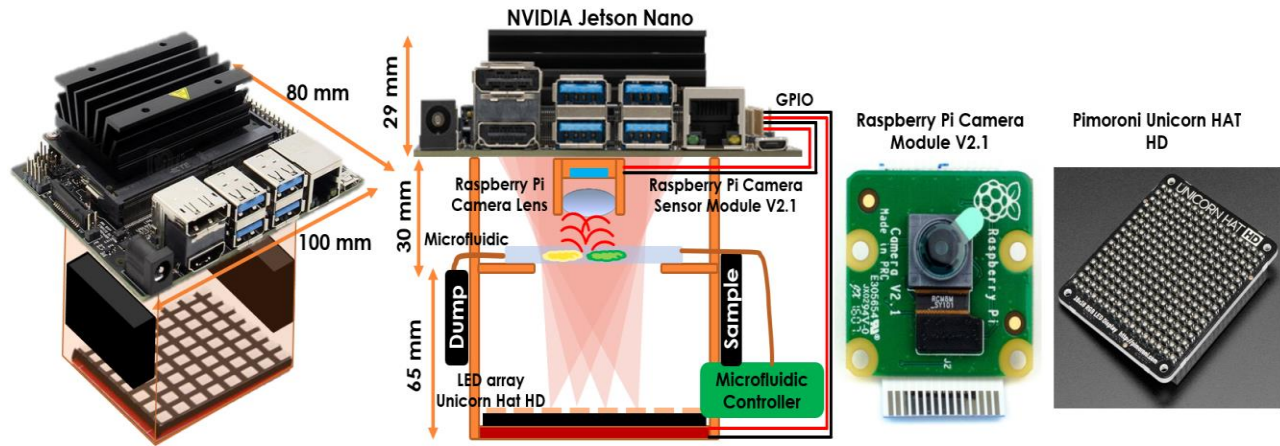


Figure 1: Components that make up our portable inferencing device assembly, showing experimental set up with typical dimensions of hardware.

A Raspberry Pi V2.1 (without IR filter) camera module (8 megapixels, $1.12\mu\text{m}$ pixel size) with a 3-mm focal-length lens (0.15 NA) was positioned to achieve $\sim 1.5\times$ magnification. Spatial frequency overlap of $\sim 70\%$ will be obtained by placing the Unicorn HAT HD (16×16) LED array (3.3mm pitch [16]) placed 65mm (to provide 0.4 NA) below the sample stage. This results in a larger synthetic NA of 0.55. The control and computation are performed through a NVIDIA Jetson Nano board (embedded AI computing platform), Figure 2.

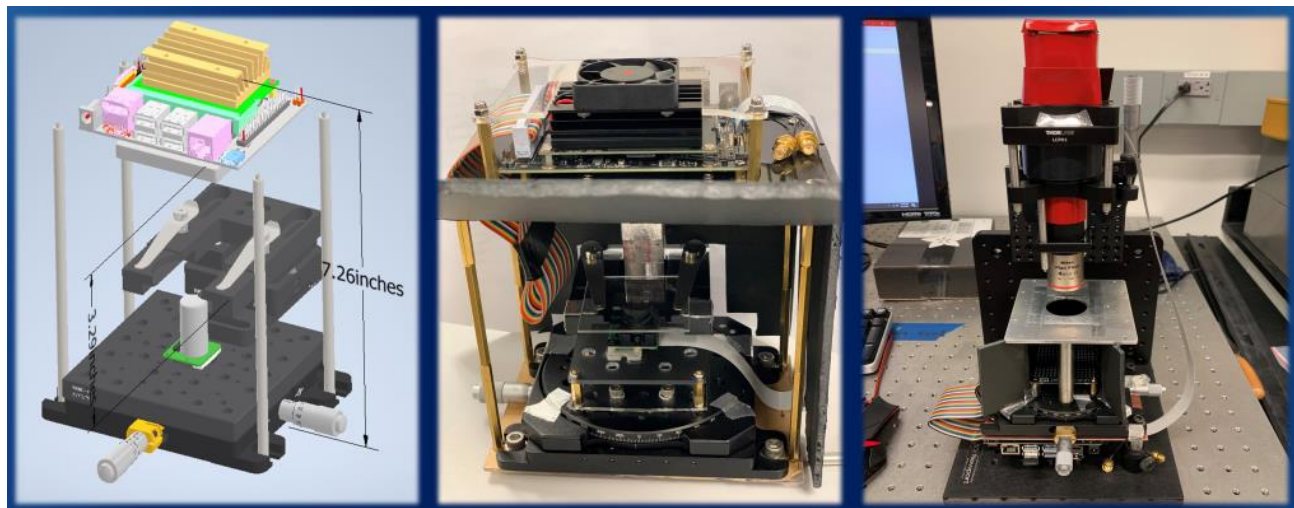


Figure 2. The NVIDIA Jetson Nano board enables the development of inferencing deep learning pipeline to provide near real-time monitoring.

The device, by itself, can be operated in the normal mode that acquires data and performs intensity, 3D phase, using the traditional FPM method with no deep learning (DL) capability. The normal mode is used to provide data for the DL mode to perform fresh training, semi-training, testing, or inferencing. The NVIDIA Jetson Nano board enables the development of inferencing deep learning pipeline to help provide near real-time monitoring. The DL model

automatically detects if the sample is out-of-focus and then mechanically repositions the sample to the focus plane using a translation stage or by digitally translating it into the focus position using DL. To implement near real-time monitoring, a DL model is used to speed up the DL-based FPM reconstruction by using 1 in 10 of the LED's (reduce data acquisition time) and the fast reconstruction ability of inferencing model. This will enhance the overall throughput of the system.

3. EXPERIMENTAL RESULTS

The data and FPM algorithm developed in [17] is used with appropriate modifications to the Python language codes for compatibility with the implemented hardware. By digitally calibrating the LED's position on each single illumination (using dual annealing method [18]) before stitching its NA-based filtered Fourier transform in global Fourier domain, artifacts have been significantly reduced especially for near-edge regions [19]. Figure 3 shows one reconstructed near-edge region. Then, the USAF target resolution (shown in Figure 4) has been used to test the performance of reconstruction. At this point, only bright field images have been used for reconstruction due to the heavy noise on dark field images that would be investigated in the next step.

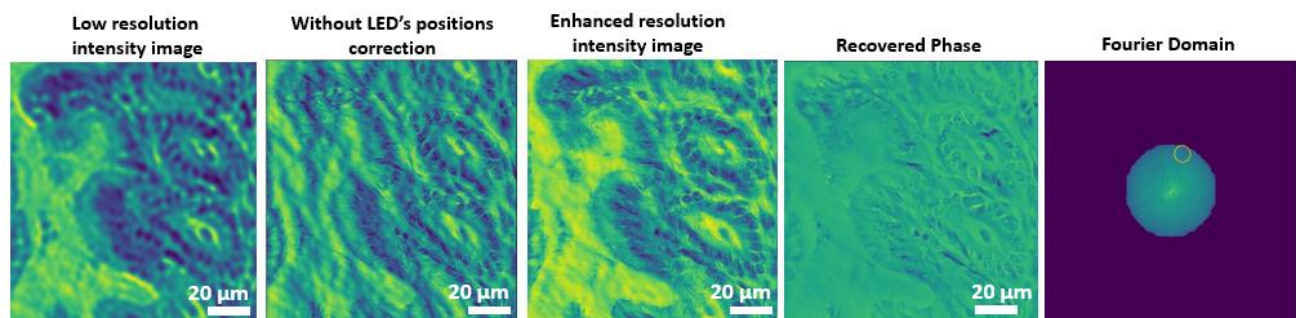


Figure 3. Results of the developed FPM reconstruction on dataset provided at [17].

Simultaneously, a computational imaging data driven model using deep convolutional neural network is developed to reconstruct high- resolution images using 1 in 10 of the LED's and having inferencing capability. Conditional generative adversarial network [20] was developed to train how to match low-resolution intensity images to high-resolution intensity/phase images. Then, its generator model (modified Unet architecture) is used as an inferencing model on NVIDIA Jetson Nano board. Cellular data set [21, 22] is used to test the inferencing model since currently we are still collecting our own dataset. 80×80 pixel-square patch of low-resolution intensity images in each region was fed into the trained model as an input to predict the phase image (320×320 pixel-square) as an output. Those phase images then were blended for a larger FOV based on vertical and horizontal coordinates. In fact, in inferencing mode, our generator model predicted ~ 20 patches/s which could be improved by deducing or modifying model. However, the quality of reconstruction needs to be taken care first.

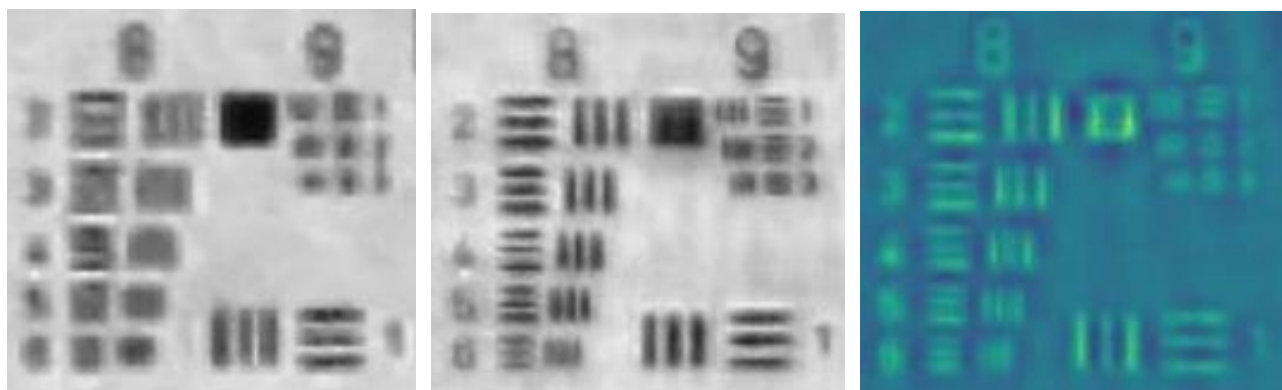


Figure 4. USAF resolution target: (left) low resolution intensity image (middle) reconstructed intensity image, (c) reconstructed phase images.

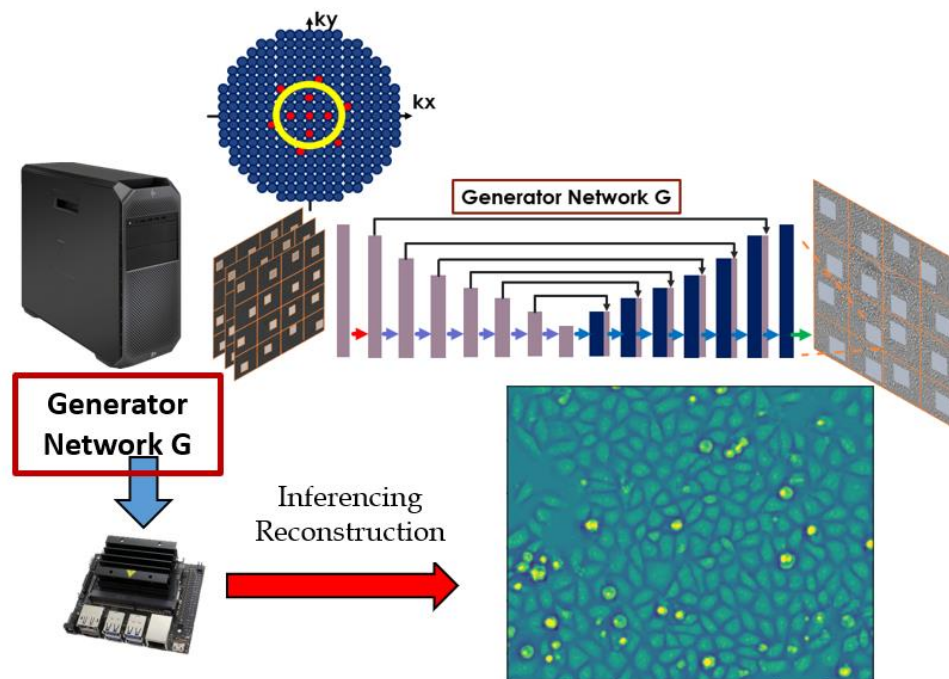


Figure 5. Deep learning framework for FPM inferencing. The generator network was trained under conditional generative adversarial architecture to predict the phase map. The trained generator model then was loaded on NVIDIA Jetson Nano board to perform inferencing.

4. CONCLUSIONS AND FUTURE WORK

The first implementation of a proof-of-concept, compact, portable FPM to perform wide field-of-view, high spatial resolution imaging has been demonstrated. FPM has proved to be an advantageous methodology to archive the high space-bandwidth-product enabling developments of novel microscope payloads on future planetary landers. Based on the current hardware set-up, it was found that fine and stable Fourier ptychographic algorithms is crucial for success since it provides the data for the learning process for reconstruction as a data-driven method and is a continuing objective for future work. Additionally, future work will focus on integrating a microfluidic device for examining liquid samples. There is a trade-off between speed of reconstruction and field-of-view area, since only sub-patches are reconstructed at a time, the model's architecture will be modified to take this into account.

5. ACKNOWLEDGEMENTS

We thank NASA Head Quarters FY20 Center Innovation Fund for supporting this work. We also thank the Computational Imaging Systems Laboratory, Boston University for the data and open source resources.

REFERENCES

- [1] Ghosh, K. K., L. D. Burns, E. D. Cocker, A. Nimmerjahn, Y. Ziv, A. El Gamal, and M. J. Schnitzer. (2011) "Miniaturized integration of a fluorescence microscope." *Nature Methods* 8: 871–878.
- [2] Swalwell, J. E., F. Ribalet, and E. V. Armbrust. (2011) "Seaflow: A novel underway flow-cytometer for continuous observations of phytoplankton in the ocean." *Limnology and Oceanography: Methods* 9(10): 466–477.
- [3] Edgett, K. S., R. A. Yingst, M. A. Ravine, M. A. Caplinger, J. N. Maki, F. T. Ghaemi, J. Schaffner, *et al.* (2012) "Curiosity's Mars Hand Lens Imager (MAHLI) Investigation." *Space Science Reviews* 170(1-4): 259–317.

- [4] Thomas, N., B. S. Luthi, S. F. Hviid, H. U. Keller, W. J. Markiewicz, T. Blumchen, A. T. Basilevsky, *et al.* (2004) "The Microscope for Beagle 2." *Planetary and Space Science* 52(9): 853–866.
- [5] Bibring, J. P., P. Lamy, Y. Langevin, A. Soufflot, M. Berthe, J. Borg, F. Poulet, and S. Mottola. (2007a) "CIVA." *Space Science Reviews* 128(1-4): 397–412.
- [6] Bibring, J. P., H. Rosenbauer, H. Boehnhardt, S. Ulamec, J. Biele, S. Espinasse, B. Feuerbacher, *et al.* (2007b) "The Rosetta Lander ("Philae") investigations." *Space Science Reviews* 128(1-4): 205–220.
- [7] Korablev, O. I., A. V. Bondarenko, I. V. Dokuchaev, A. Y. Ivanov, O. E. Kozlov, V. A. Kottsov, A. B. Kiselev, J. P. Bibring, and J. J. Fourmond. (2010) "Microscope spectrometer for the Phobos-Grunt mission." *Solar System Research* 44(5): 403–408.
- [8] Chan, M. A., Hinman, N. W., Potter-McIntyre, S. L., Schubert, K. E., Gillams, R. J., Awramik, S. M., *et al.* (2019) "Deciphering biosignatures in planetary contexts." *Astrobiology*, 19(9), 1075-1102.
- [9] Yingst, R. A., Edgett, K. S., Kennedy, M. R., Krezoski, G. M., McBride, M. J., *et al.* (2016) "MAHLI on Mars: lessons learned operating a geoscience camera on a landed payload robotic arm." *Geoscientific Instrumentation, Methods and Data Systems*, 5(1), 205-217.
- [10] Bhartia, R., E. C. Salas, W. F. Hug, R. D. Reid, A. L. Lane, K. J. Edwards, and K. H. Nealson. (2010) "Label-free bacterial imaging with deep-UV-laser-induced native fluorescence." *Applied and Environmental Microbiology* 76(21): 7231–7237.
- [11] Santibáñez, P. A., J. R. McConnell, and J. C. Prisco. (2016) "A flow cytometric method to measure prokaryotic records in ice cores: An example from the West Antarctic Ice Sheet Divide drilling site." *Journal of Glaciology* 62(234): 655–673.
- [12] Hecht, M. H., J. Marshall, W. T. Pike, U. Staufer, D. Blaney, D. Braendlin, S. Gautsch, *et al.* (2008) "Microscopy capabilities of the Microscopy, Electrochemistry, and Conductivity Analyzer." *Journal of Geophysical Research: Planets* 113(E3). doi:10.1029/2008je003077.
- [13] Gaskin, J. A., T. Abbott, S. Medley, K. Patty, D. Gregory, K. Thaisen, B. Ramsey, *et al.* (2010) "Miniaturized scanning electron microscope for in-situ planetary studies." In *Earth and Space 2010: Engineering, Science, Construction, and Operations in Challenging Environments*, edited by G. Song and R. B. Malla: American Society of Civil Engineers, p. 1246–1257.
- [14] Europa Lander Study 2016 Report, Europa Lander Mission, JPL D-97667
- [15] Summons, R. E., J. P. Amend, D. Bish, R. Buick, G. D. Cody, D. J. Des Marais, G. Dromart, *et al.* (2011) "Preservation of Martian organic and environmental records: Final Report of the Mars Biosignature Working Group." *Astrobiology* 11(2): 157–181.
- [16] Tomas A., Regina E., Andrew R. H., Laura W., and Pavan C. K. "Low-cost, sub-micron resolution, wide-field computational microscopy using opensource hardware." *Scientific reports* 9, no. 1 (2019): 1-12.
- [17] Lei T., Xiao L., Kannan R., and L. Waller, "Multiplexed coded illumination for Fourier Ptychography with an LED array microscope," *Biomed. Opt. Express* 5, 2376-2389 (2014).
- [18] Xiang Y., Sun D.Y., Fan W., Gong X.G., "Generalized Simulated Annealing Algorithm and Its Application to the Thomson Model", *Physics Letters A*, 233, 216-220 (1997)
- [19] Ao Z., Wei W., Ni C., Edmund Y. L., Byoung-ho L., and Guohaiz S., "Fast and robust misalignment correction of Fourier ptychographic microscopy for full field of view reconstruction," *Opt. Express* 26, 23661-23674 (2018).
- [20] Mehdi M., Osindero S., "Conditional Generative Adversarial Nets", printed at <https://arxiv.org/abs/1411.1784v1>
- [21] Thanh N., Yujia X., Yunzhe L., Lei T., and George N., "Deep learning approach for Fourier ptychography microscopy," *Opt. Express* 26, 26470-26484 (2018)
- [22] Lei T., Ziji L., Li-Hao Y., Michael C., Jingshan Z., and Laura W., "Computational illumination for high-speed in vitro Fourier ptychographic microscopy," *Optica* 2, 904-911 (2015)